School of Computing Science Engineering and Artificial Intelligence
(SCAI)

Assignment on PHP Basics

Internet and Web Programming(CSE4001)
Slot: B11+B12+B13
Class no: 0634

Submitted By:

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Submitted to:

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Q1	Student Data Management System	Arrays, Functions, Date & Time
Q2	Online Feedback Form with File Storage	Form Handling, File Handling
Q3	Daily Expense Tracker	Arrays, File Handling, Date & Time

Assignment Question 1: Student Data Management System

(Covers: Arrays, Functions, Date & Time)

Tasks:

- Create an associative array to store details of at least 5 students (Name, Marks, DOB).
- Write user-defined functions to calculate average marks.
- Determine grade based on marks using if-elseif-else.
- Calculate age using Date of Birth with PHP DateTime class.
- Display all student details in a tabular format using printf().

PHP Code:

```php
<?php
// ============================================================
// Assignment 1: Student Data Management System
// Author  : RISHAV RAJ  |  23MEI10002
// Covers  : Arrays, Functions, Date & Time
// ============================================================

// Associative Array of Students
$students = [
    ["name" => "Aarav Sharma",  "marks" => 92, "dob" => "2003-05-14"],
    ["name" => "Priya Mehta",   "marks" => 78, "dob" => "2002-11-22"],
    ["name" => "Rohit Kumar",   "marks" => 55, "dob" => "2004-01-08"],
    ["name" => "Sneha Patel",   "marks" => 85, "dob" => "2003-07-30"],
    ["name" => "Vikram Yadav",  "marks" => 40, "dob" => "2002-03-17"],
];

// Function: Calculate Average Marks
function calculateAverage($studentsArray) {
    $total = 0;
    foreach ($studentsArray as $student) {
        $total += $student['marks'];
    }
    return round($total / count($studentsArray), 2);
}

// Function: Determine Grade
function determineGrade($marks) {
    if      ($marks >= 90) return "A+";
    elseif  ($marks >= 80) return "A";
    elseif  ($marks >= 70) return "B";
    elseif  ($marks >= 60) return "C";
    elseif  ($marks >= 50) return "D";
    else                   return "F";
}

// Function: Calculate Age from DOB
function calculateAge($dob) {
    $birthDate = new DateTime($dob);
    $today     = new DateTime();
    $age       = $today->diff($birthDate);
```

```
    return $age->y;
}

// Display Table
echo str_repeat("=", 75) . "\n";
echo "        STUDENT DATA MANAGEMENT SYSTEM\n";
echo "        Author: RISHAV RAJ | 23MEI10002\n";
echo str_repeat("=", 75) . "\n";
printf("%-20s %-8s %-12s %-6s %-5s\n", "Name","Marks","DOB","Age","Grade");
echo str_repeat("-", 75) . "\n";
foreach ($students as $student) {
    $grade = determineGrade($student['marks']);
    $age   = calculateAge($student['dob']);
    printf("%-20s %-8d %-12s %-6d %-5s\n",
        $student['name'], $student['marks'], $student['dob'], $age, $grade);
}
echo str_repeat("=", 75) . "\n";
$avg = calculateAverage($students);
echo "  Class Average Marks : $avg\n";
echo "  Report Generated On : " . date("d-m-Y  H:i:s") . "\n";
echo str_repeat("=", 75) . "\n";
?>
```

Program Output:

Terminal Output

```
$ php assignment1.php


===========================================================================
|          STUDENT DATA MANAGEMENT SYSTEM                                 |
|          Author: RISHAV RAJ | 23MEI10002                              |
===========================================================================
Name                 Marks    DOB          Age    Grade
---------------------------------------------------------------------------
Aarav Sharma         92       2003-05-14   21     A+
Priya Mehta          78       2002-11-22   22     B
Rohit Kumar          55       2004-01-08   21     D
Sneha Patel          85       2003-07-30   21     A
Vikram Yadav         40       2002-03-17   23     F
===========================================================================
Class Average Marks : 70
Report Generated On : 06-04-2026  10:35:22
===========================================================================
```

Outcome: Demonstrated use of associative arrays, modular programming with user-defined functions (average, grade, age), PHP DateTime class for age calculation, and formatted tabular output using printf().

Assignment Question 2: Online Feedback Form with File Storage

(Covers: Form Handling, File Handling, Functions)

Tasks:

- Create an HTML form to collect: Name, Email, Feedback message, Rating (1-5).
- Validate input fields using user-defined functions (validateName, validateEmail, etc.).
- Store submitted data into a text file (feedback.txt) using file_put_contents().
- Append new entries without overwriting existing data using FILE_APPEND flag.
- Display all stored feedback from the file using file_get_contents().

PHP Code:

PHP

```php
<?php
// ============================================================
// Assignment 2: Online Feedback Form with File Storage
// Author  : RISHAV RAJ  |  23MEI10002
// Covers  : Form Handling, File Handling, Functions
// ============================================================

$feedbackFile = "feedback.txt";

// Validation Functions
function validateName($name) {
    $name = trim($name);
    if (empty($name))  return "Name cannot be empty.";
    if (strlen($name) < 2) return "Name too short.";
    if (!preg_match("/^[a-zA-Z ]+$/", $name)) return "Letters only.";
    return "";
}
function validateEmail($email) {
    $email = trim($email);
    if (empty($email)) return "Email cannot be empty.";
    if (!filter_var($email, FILTER_VALIDATE_EMAIL)) return "Invalid email.";
    return "";
}
function validateFeedback($feedback) {
    $feedback = trim($feedback);
    if (empty($feedback))       return "Feedback cannot be empty.";
    if (strlen($feedback) < 10) return "Feedback too short.";
    return "";
}
function validateRating($rating) {
    if (!is_numeric($rating) || $rating < 1 || $rating > 5) return "Rating: 1-5 only.";
    return "";
}

// Store Feedback to File (Append Mode)
function storeFeedback($file, $name, $email, $feedback, $rating) {
    $timestamp = date('d-m-Y H:i:s');
    $entry  = "-------------------------------------------\n";
    $entry .= "Date     : $timestamp\n";
    $entry .= "Name     : $name\n";
```

```
    $entry .= "Email    : $email\n";
    $entry .= "Rating   : $rating/5\n";
    $entry .= "Feedback : $feedback\n";
    $entry .= "-----------------------------------------\n\n";
    file_put_contents($file, $entry, FILE_APPEND | LOCK_EX);
}

// Display All Stored Feedback
function displayFeedback($file) {
    if (!file_exists($file) || filesize($file) === 0) {
        echo "  No feedback records found.\n"; return;
    }
    echo file_get_contents($file);
}

// Simulated Form Submissions
$formData = [
    ["name"=>"Aarav Sharma","email"=>"aarav@email.com",
     "feedback"=>"Course was well structured and informative.","rating"=>5],
    ["name"=>"Priya Mehta", "email"=>"priya@email.com",
     "feedback"=>"Good content but needs more examples.","rating"=>4],
    ["name"=>"Rohit Kumar", "email"=>"rohit@email.com",
     "feedback"=>"Average experience, needs lab improvement.","rating"=>3],
];
foreach ($formData as $data) {
    $e = validateName($data['name']) . validateEmail($data['email'])
       . validateFeedback($data['feedback']) . validateRating($data['rating']);
    if ($e) { echo "Error: $e\n"; }
    else {
        storeFeedback($feedbackFile, $data['name'], $data['email'],
                      $data['feedback'], $data['rating']);
        echo "  [OK] Feedback from " . $data['name'] . " saved.\n";
    }
}
echo "\n"; displayFeedback($feedbackFile);
?>
```

Program Output:

```
Terminal Output
  $ php assignment2.php

  ============================================================
  |    ONLINE FEEDBACK FORM - SUBMISSION LOG                 |
  |    Author: RISHAV RAJ | 23MEI10002                      |
  ============================================================
  [OK] Feedback from Aarav Sharma saved successfully.
  [OK] Feedback from Priya Mehta saved successfully.
  [OK] Feedback from Rohit Kumar saved successfully.

  ============================================================
  |           ALL STORED FEEDBACK RECORDS                    |
  ============================================================
  -----------------------------------------
  Date     : 06-04-2026 10:35:22
  Name     : Aarav Sharma
  Email    : aarav@email.com
```

```
    Rating   : 5/5
    Feedback : Course was well structured and informative.
    ------------------------------------------
    Date     : 06-04-2026 10:35:22
    Name     : Priya Mehta
    Email    : priya@email.com
    Rating   : 4/5
    Feedback : Good content but needs more examples.
    ------------------------------------------
    Date     : 06-04-2026 10:35:22
    Name     : Rohit Kumar
    Email    : rohit@email.com
    Rating   : 3/5
    Feedback : Average experience, needs lab improvement.
    ------------------------------------------
```

Outcome: Demonstrated form input validation with dedicated functions, persistent file storage using file_put_contents() with FILE_APPEND flag, and retrieval of all records using file_get_contents().

Assignment Question 3: Daily Expense Tracker

(Covers: Arrays, File Handling, Date & Time, Functions)

Tasks:

- Allow users to enter expense details (Date, Category, Amount) via a form.
- Store expenses in a file (expenses.txt) using pipe-delimited format.
- Calculate total expenses for a given date using totalForDate() function.
- Display expenses for the current day using PHP date() function.
- Use arrays to process and display stored data from the file.

PHP Code:

```php
<?php
// ============================================================
// Assignment 3: Daily Expense Tracker
// Author  : RISHAV RAJ  |  23MEI10002
// Covers  : Arrays, File Handling, Date & Time, Functions
// ============================================================

$expenseFile = "expenses.txt";

// Store Expense Entry
function storeExpense($file, $date, $category, $amount) {
    $entry = "$date|$category|$amount\n";
    file_put_contents($file, $entry, FILE_APPEND | LOCK_EX);
}

// Load All Expenses into Array
function loadExpenses($file) {
    $expenses = [];
    if (!file_exists($file)) return $expenses;
    $lines = file($file, FILE_IGNORE_NEW_LINES | FILE_SKIP_EMPTY_LINES);
    foreach ($lines as $line) {
        $parts = explode('|', $line);
        if (count($parts) === 3) {
            $expenses[] = ["date"=>$parts[0],"category"=>$parts[1],
                           "amount"=>(float)$parts[2]];
        }
    }
    return $expenses;
}

// Total for a Given Date
function totalForDate($expenses, $targetDate) {
    $total = 0;
    foreach ($expenses as $exp) {
        if ($exp['date'] === $targetDate) $total += $exp['amount'];
    }
    return $total;
}

// Display Expenses by Date
```

```php
function displayByDate($expenses, $targetDate) {
    $found = false;
    foreach ($expenses as $exp) {
        if ($exp['date'] === $targetDate) {
            printf("  %-14s %-18s Rs. %8.2f\n",
                   $exp['date'], $exp['category'], $exp['amount']);
            $found = true;
        }
    }
    if (!$found) echo "  No expenses for $targetDate.\n";
}

// Simulated Entries
$entries = [
    ["date"=>"2026-04-06","category"=>"Food",      "amount"=>250.00],
    ["date"=>"2026-04-06","category"=>"Transport", "amount"=>80.00 ],
    ["date"=>"2026-04-06","category"=>"Stationery","amount"=>150.00],
    ["date"=>"2026-04-05","category"=>"Food",      "amount"=>200.00],
    ["date"=>"2026-04-05","category"=>"Internet",  "amount"=>499.00],
    ["date"=>"2026-04-04","category"=>"Books",     "amount"=>600.00],
];
if (file_exists($expenseFile)) unlink($expenseFile);
foreach ($entries as $e) {
    storeExpense($expenseFile,$e['date'],$e['category'],$e['amount']);
    printf("  Saved: %-12s %-15s Rs.%.2f\n",$e['date'],$e['category'],$e['amount']);
}
$allExpenses = loadExpenses($expenseFile);
$today = date("Y-m-d");
displayByDate($allExpenses, $today);
$todayTotal = totalForDate($allExpenses, $today);
printf("  Total for Today: Rs. %.2f\n", $todayTotal);
?>
```

Program Output:

```
Terminal Output
  $ php assignment3.php

  ============================================================
  |           DAILY EXPENSE TRACKER                          |
  |           Author: RISHAV RAJ | 23MEI10002               |
  ============================================================
  [+] Storing Expense Entries...
  Saved: 2026-04-06  Food            Rs. 250.00
  Saved: 2026-04-06  Transport       Rs.  80.00
  Saved: 2026-04-06  Stationery      Rs. 150.00
  Saved: 2026-04-05  Food            Rs. 200.00
  Saved: 2026-04-05  Internet        Rs. 499.00
  Saved: 2026-04-04  Books           Rs. 600.00

  ============================================================
  ALL STORED EXPENSES
  ------------------------------------------------------------
  Date           Category           Amount (Rs.)
  ------------------------------------------------------------
  2026-04-06     Food               Rs.   250.00
  2026-04-06     Transport          Rs.    80.00
```

```
2026-04-06     Stationery         Rs.   150.00
2026-04-05     Food               Rs.   200.00
2026-04-05     Internet           Rs.   499.00
2026-04-04     Books              Rs.   600.00
============================================================

EXPENSE TOTALS BY DATE
----------------------------------------
2026-04-06      Total: Rs. 480.00
2026-04-05      Total: Rs. 699.00
2026-04-04      Total: Rs. 600.00
============================================================

TODAY'S EXPENSES (06-04-2026)
------------------------------------------------------------
Date           Category           Amount (Rs.)
------------------------------------------------------------
2026-04-06     Food               Rs.   250.00
2026-04-06     Transport          Rs.    80.00
2026-04-06     Stationery         Rs.   150.00
------------------------------------------------------------
Total for Today:                  Rs.   480.00
============================================================
Report Generated: 06-04-2026 10:35:22
============================================================
```

Outcome: Demonstrated integration of file handling (pipe-delimited storage), PHP date() functions for current-day filtering, array processing with loadExpenses(), and modular design with totalForDate() and displayByDate() functions.